Charlie Sands

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Education

Olin College of Engineering, B.S. Electrical and Computer Engineering | 4.00/4.00 GPA **Sep 2024 - May 2027**

- Relevant coursework: Multivariable Calculus, Signals and Systems, Microelectronics, Sensing and Instrumentation, E & M, Microwave Engineering, Antenna Theory, Differential Equations, Computer Architecture

Professional Work Experience

RF Design Engineering Intern | Space Exploration Technologies Corp. (Starlink) **May 2026 – Aug 2026**

- Designed new extremely low phase-noise VCOs operating between C and K-band for up and down converters
- Developed new automated manufacturing techniqes for enabling mass production of VCOs faster than vendors
- Production cut-in for new VCOs targeted for January 2027, pending qualification campaign results and supply chain

RF & Antenna Design Engineer | Mithril Technologies Inc. **Sep 2025 – Present**

- Created a modeling algorithm for a state-of-the-art electrostatically reconfigurable parabolic antenna reflector
- Developed instrumentation and test equipment to accurately profile performance of parabolic antenna prototypes
- Designed, manufactured and tested a prototype proof-of-concept antenna reflector, gathering preliminary results
- Wrote a first-author conference paper detailing the design, proposed implementation techniques and simulation results, accepted for presentation at *IEEE AP-S International Conference on Antennas and Propagation* in July 2026
- Part of team that wrote a conference paper and SBIR grant application outlining novel mission CONOPS enabled by new antenna design, paper presented at *AIAA ASCEND* in May 2026 and SBIR grant awarded in June 2026

RF Design Engineering Intern | Astranis Space Technologies Corp. **May 2025 – Aug 2025**

- Designed, manufactured and tested failure tolerant waveguide power combiner for military Ka-band geostationary communications satellites, reaching over 98% power combining efficiency across a >1 GHz bandwidth
- Redesigned and reworked X-band up and down converters planned for use in software defined radio transmit and receive pipelines on geostationary communications satellites, reducing undesirable spur levels by over 80%
- Designed and built a software defined radio-based EMI/EMC calibrator for less than 10% the cost of commercial equipment that is used for test equipment calibration prior to communications satellite compliance testing
- Specified components and redesigned circuit boards for Ka-band TTC modem transmit and receive frequency converters, significantly reducing in band spurs, increasing conversion gain and other figures of merit

RF Design & Avionics Engineering Lead | Olin College Rocketry **Sep 2024 – Aug 2025**

- Developed a long-range 4+ watt half-duplex L-band radio telemetry transceiver with a custom signal chain design

Research & Personal Projects

L-Band Inductorless 180nm BiCMOS LNA | *RFIC, Analog IC Design, Klayout, NGSpice* **Oct 2026 – Present**

- Leading design of an open source LNA operating between 0.9 – 2 GHz on a Global Foundries 180 nm CMOS process
- Performing schematic, layout and electromagnetic simulations for the full design ahead of tape out in early 2027

Wirelessly Powered Sensing Platform | *RF Engineering, Ansys EDT, C, SDR, PCB Design* **Sep 2024 – Sep 2025**

- Designed, simulated and manufactured a new passively powered sensing system energized by an S-band beacon, equipped with a microcomputer and sensors and capable of relaying information through backscatter communication
- Implemented systems allowing sensor data encoding on beacon sidebands for extremely low-power transmissions
- Wrote a first-author conference paper detailing the design and testing methodology of the system, presented at the *IEEE AP-S Topical Conference on Antennas and Propagation in Wireless Communications* in September 2025

GOES-19 Satellite Ground Station | *RF Engineering, Ansys EDT, C, SDR, PCB Design, Mixed Signal* **Oct 2025 – Dec 2025**

- Developed a custom LNB including amplifiers, frequency synthesizer and downconverter to receive satellite imagery

Microwave Radar System | *RF Engineering, Electrical Engineering, PCB Design, C, Python* **Sep 2022 – Jul 2023**

- Designed a S-band frequency modulated continuous wave radar system for detection and ranging of moving targets
- Manufactured and implemented custom circuit boards to connect the RFIC components that comprise the radar
- Developed custom frontend and drivers in Python and C to interface analog radar electronics to modern Linux PCs

Skills

- **Electrical:** *KiCAD*, *LTSpice, Altium, Keysight ADS, Ansys EDT*, Mixed Signal Design, RF Design, Oscilloscope, Spectrum Analyzer, VNA
- **Software:** Assembly, *C*, *C++*, *Java*, *Python*, BASH, *MATLAB*, *Git*, *GNURadio*, SDR Platforms, Linux Kernel
- **Mechanical:** *SolidWorks*, *Ansys Thermal*, Tormach CNC, Manual Mill, Manual Lathe, 3D Printing, Laser Cutting